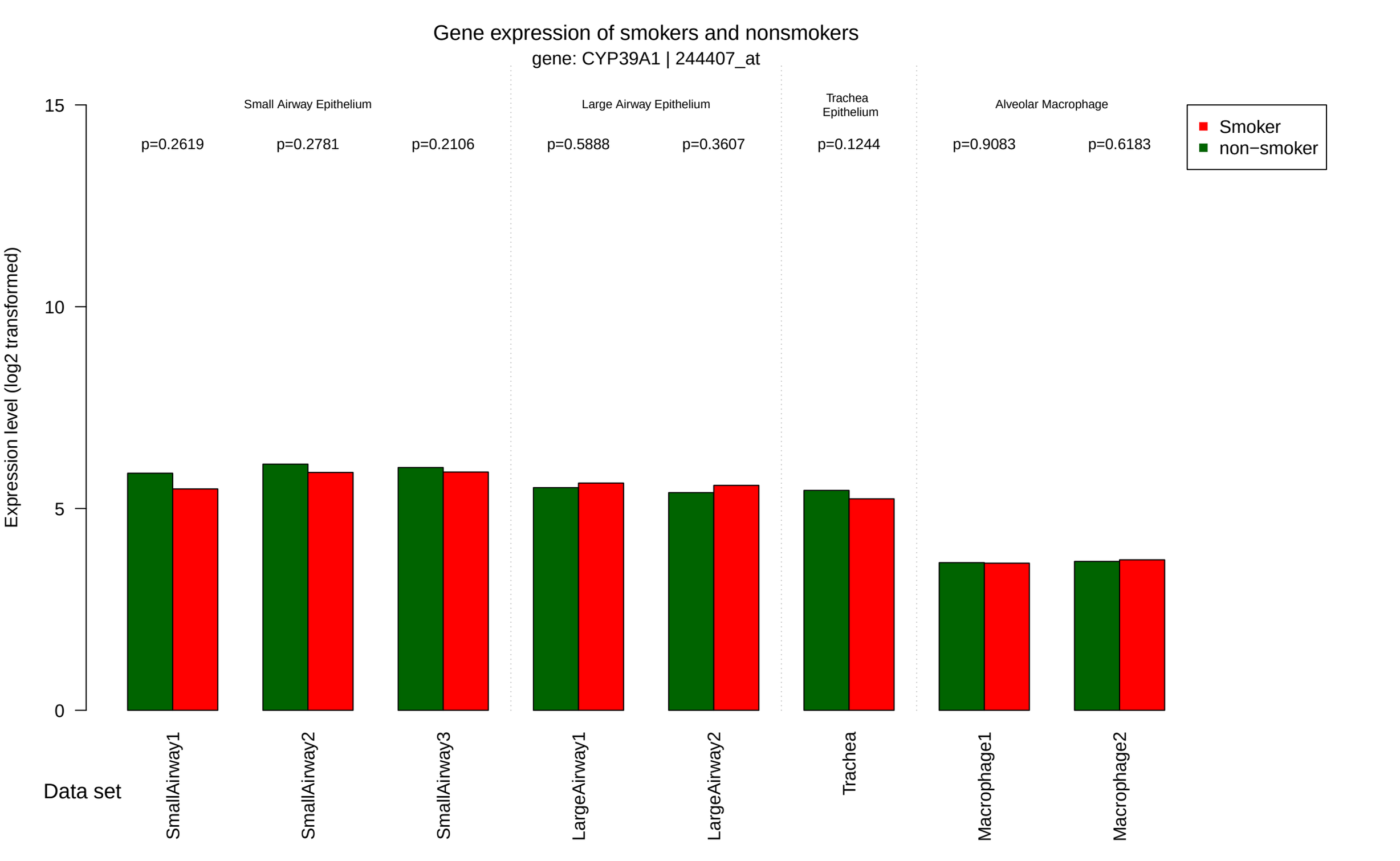
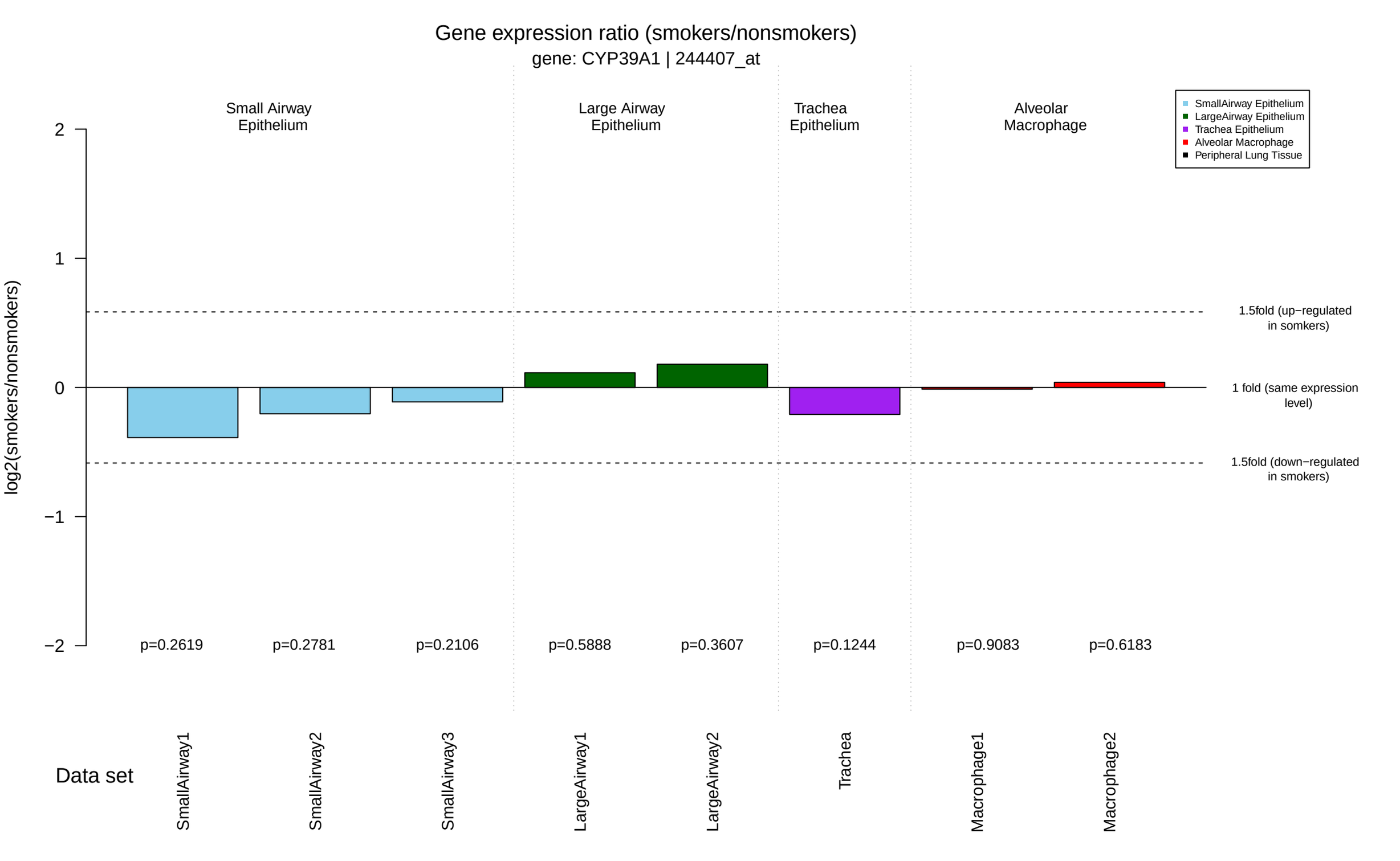
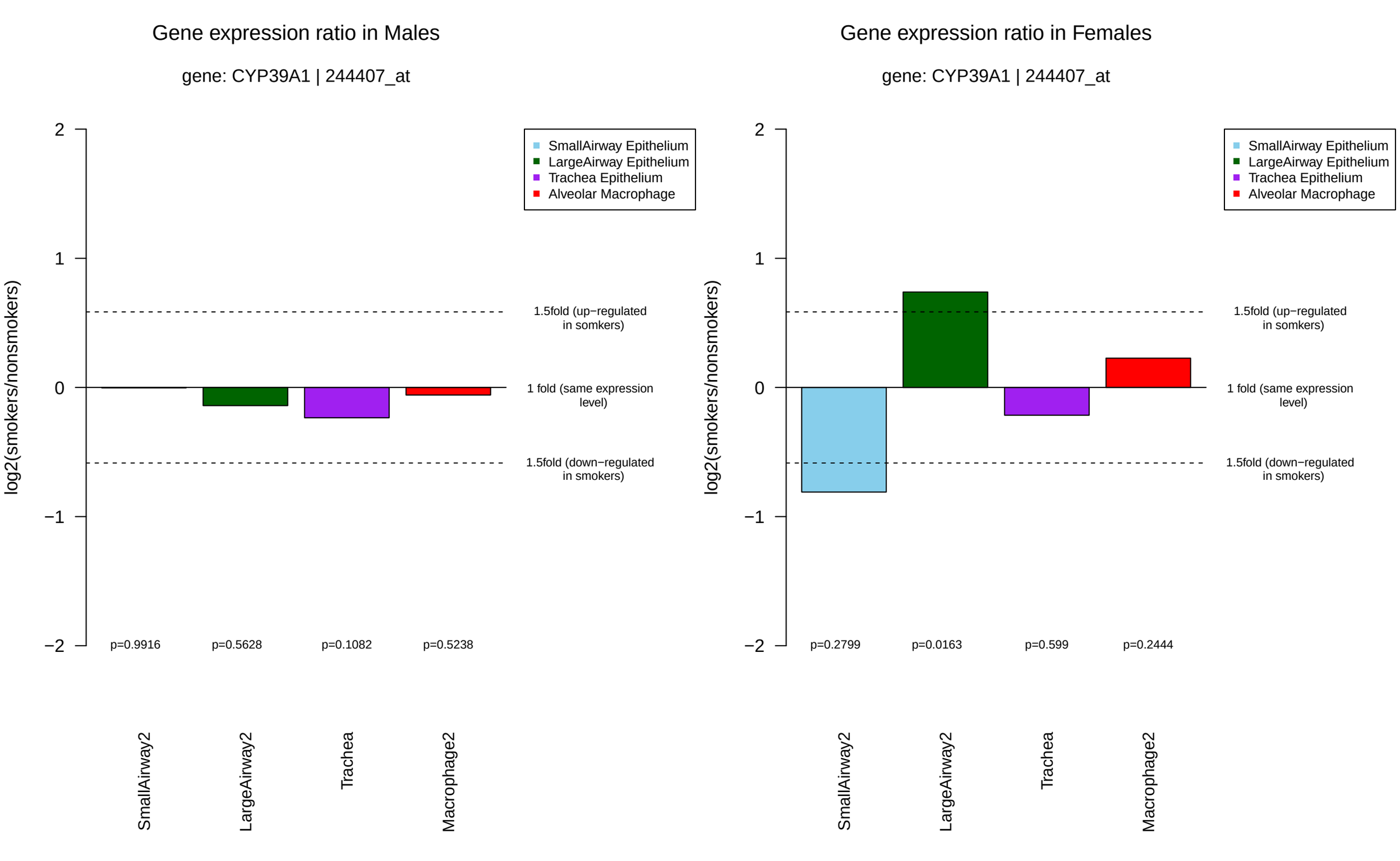
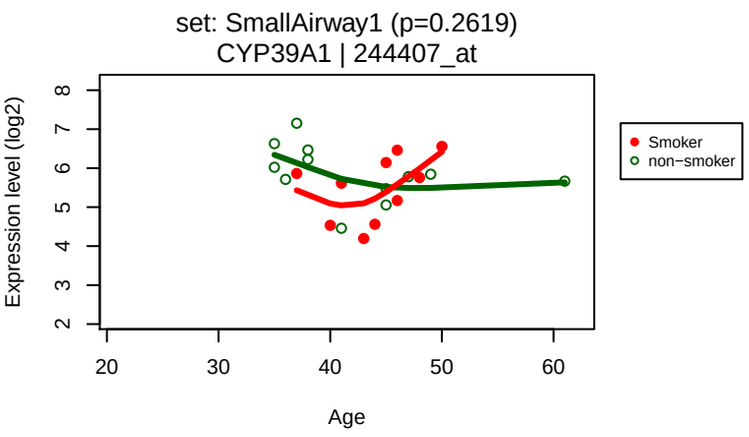
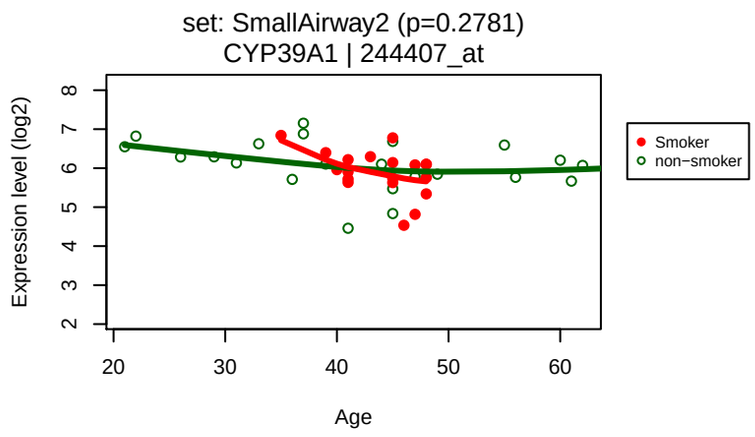
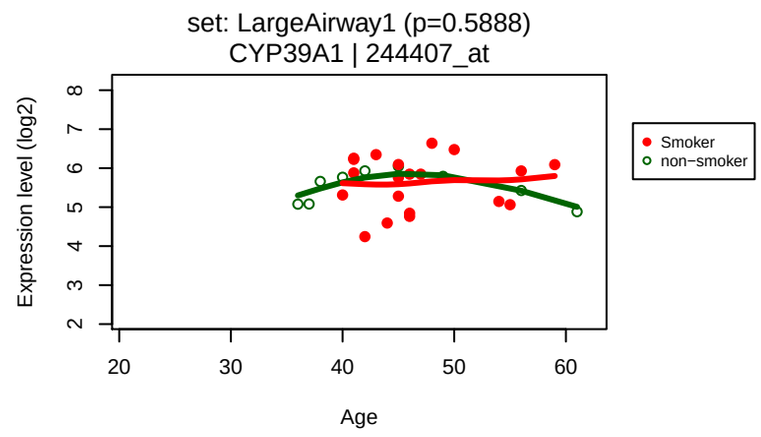

Gene expression of smokers and nonsmokers
gene: CYP39A1 | 244407_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2619
p=0.2781
p=0.2106
p=0.5888
p=0.3607
p=0.1244
p=0.9083
p=0.6183
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

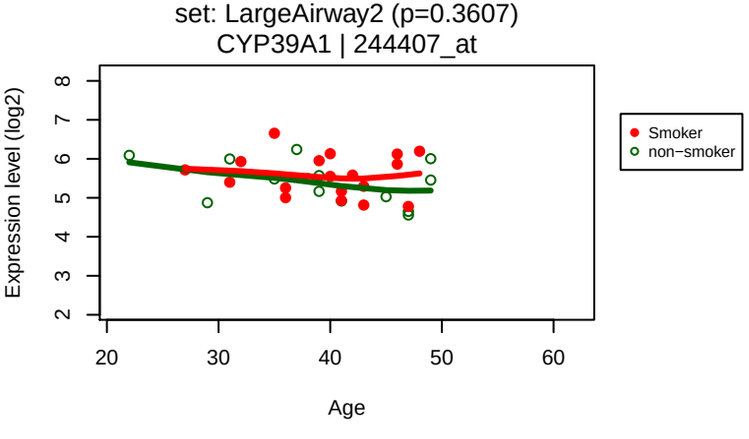

Gene expression ratio (smokers/nonsmokers)
gene: CYP39A1 | 244407_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2619
p=0.2781
p=0.2106
p=0.5888
p=0.3607
p=0.1244
p=0.9083
p=0.6183
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

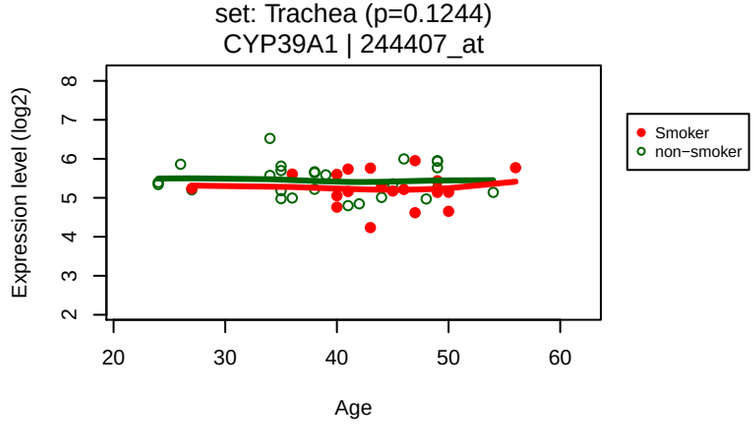

Gene expression ratio in Males
gene: CYP39A1 | 244407_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9916
p=0.5628
p=0.1082
p=0.5238
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP39A1 | 244407_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2799
p=0.0163
p=0.599
p=0.2444
SmallAirway2
LargeAirway2
Trachea
Macrophage2

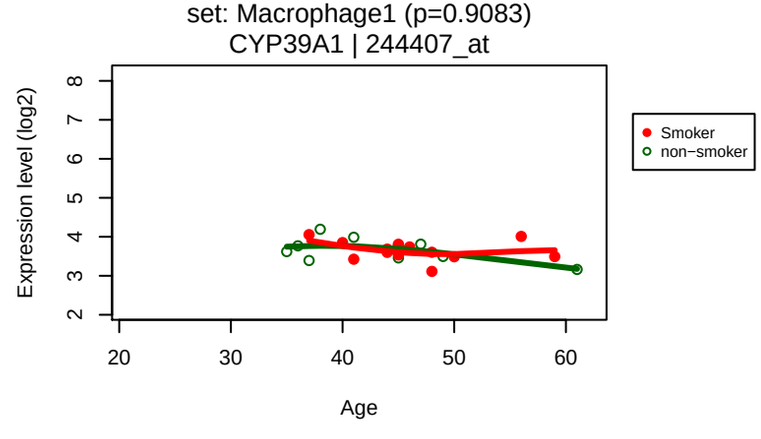

set: SmallAirway1 (p=0.2619)
CYP39A1 | 244407_at
Expression level (log2)
Age
Smoker
non−smoker

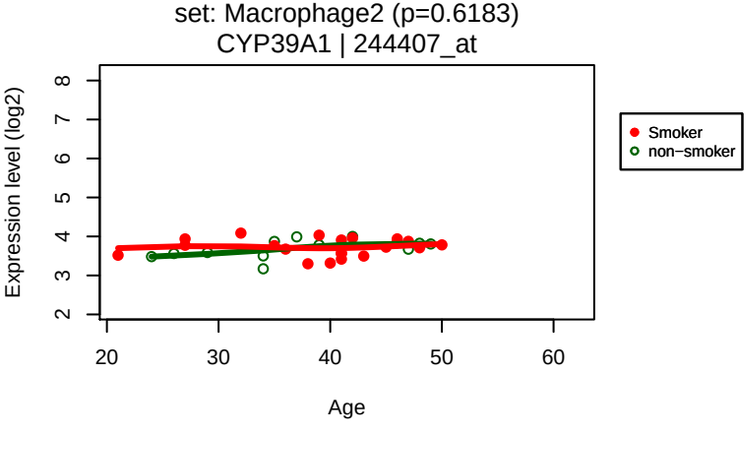

set: SmallAirway2 (p=0.2781)
CYP39A1 | 244407_at
Expression level (log2)
Age
Smoker
non−smoker

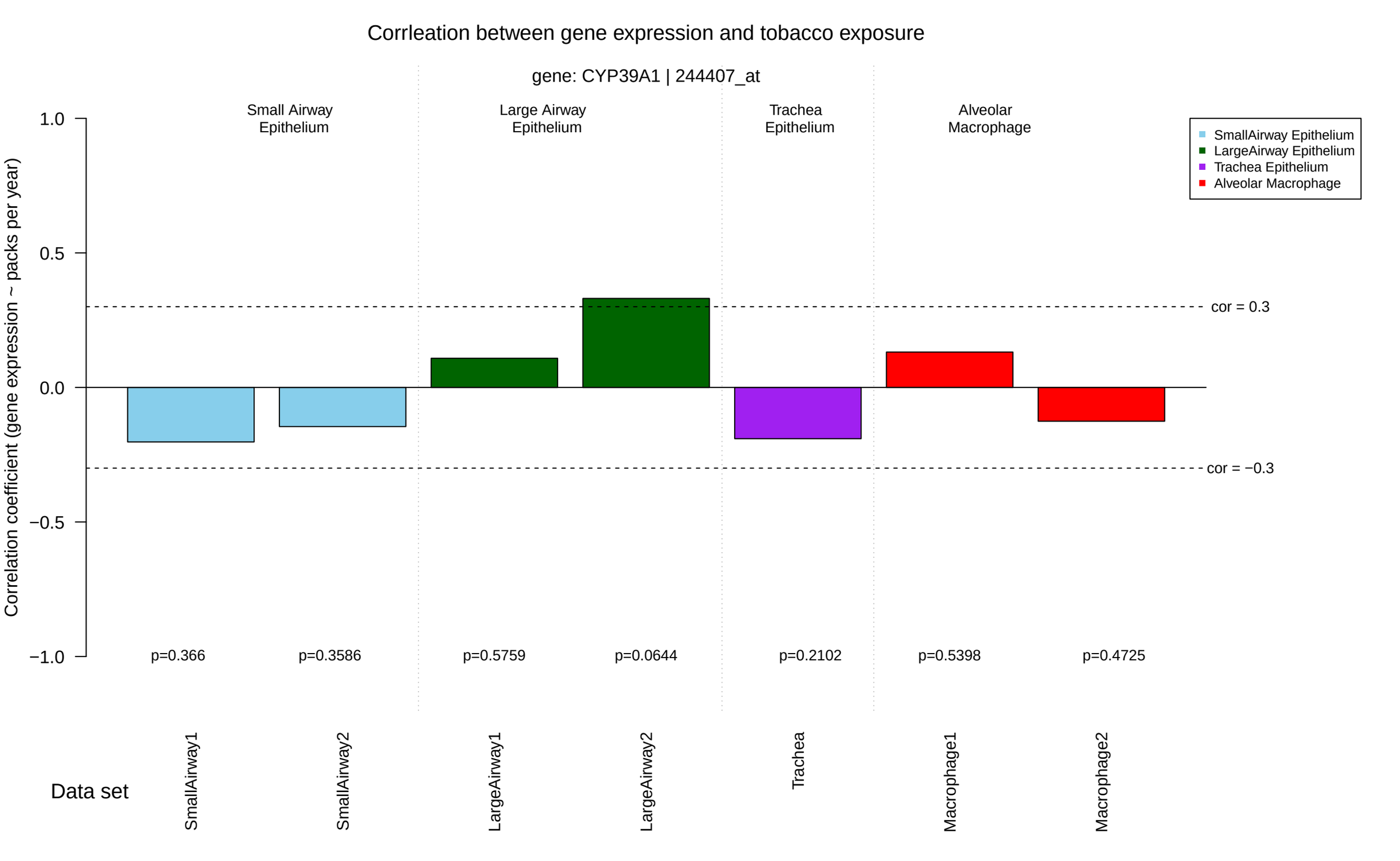

set: LargeAirway1 (p=0.5888)
CYP39A1 | 244407_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3607)
CYP39A1 | 244407_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1244)
CYP39A1 | 244407_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9083)
CYP39A1 | 244407_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6183)
CYP39A1 | 244407_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP39A1 | 244407_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.366
p=0.3586
p=0.5759
p=0.0644
p=0.2102
p=0.5398
p=0.4725
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2